

## Nova — AI Voice Assistant Project

### *An Intelligent, Voice-Enabled Web-Based Assistant*

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#### 1. Project Overview

**Nova** is a modern AI-powered voice assistant designed to perform everyday digital tasks through natural language commands. It combines **voice interaction, web automation, and intelligent responses** into a seamless browser-based experience.

The system allows users to **ask questions, execute commands, retrieve information, and automate tasks** using either voice or text input.

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#### 2. Problem Statement

Most existing assistants:

- Are platform-dependent (Google Assistant, Siri)
- Require installation or ecosystem lock-in
- Lack customization for developers/students

**Nova solves this by:**

- Running entirely in a browser
  - Being lightweight and open for customization
  - Supporting both voice and text interaction
  - Providing developer-friendly architecture
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#### 3. Solution

Nova provides:

- A **web-based AI assistant**
- **Voice recognition + speech output**
- **Command execution engine**
- **Real-time information retrieval**

Users can interact naturally, and Nova interprets commands to perform actions instantly.

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## 4. ✨ Key Features

### Intelligent Commands

- Answer general knowledge questions
- Perform calculations
- Provide study plans

### Web Automation

- Open websites (YouTube, GitHub, Google, etc.)
- Search web, YouTube, Wikipedia
- Open maps and locations

### Voice Interaction

- Speech-to-text input (browser mic)
- Text-to-speech responses (browser + system)

### Productivity Tools

- Notes & reminders
- Todo list management
- Password generation
- Unit conversion

### Utility Functions

- Dice rolling, coin flipping
  - Word counting & text transformation
  - Time & date queries
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## 5. 🏗️ System Architecture

### Backend:

- **Python + Flask**
- Handles command processing and logic

- Integrates APIs (Wikipedia, etc.)

#### Frontend:

- HTML, CSS, JavaScript
- Interactive chat interface
- Voice input via Web Speech API

#### Voice System:

- **Browser:** SpeechSynthesis (no install required)
- **Server:** pyttsx3 (offline voice support)

## 6. 📁 Project Structure

Nova\_Assistant/

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|
|
|— app.py          → Main Flask server
|— requirements.txt → Dependencies
|
|— assistant/
|   |— commands.py → Command processing logic
|   |— speak.py   → Text-to-speech
|   └— utils.py    → Helper functions
|
|— templates/
|   └— index.html   → Web UI
|
└— static/
    └— style.css    → UI styling
  
```

## 7. ⚙️ Working Flow

1. User gives input (text or voice)
  2. Input is processed in Flask backend
  3. Command is matched with predefined handlers
  4. Action is executed (search/open/respond)
  5. Output is:
    - Displayed on screen
    - Spoken via voice system
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## 8. Technologies Used

- **Python**
  - **Flask**
  - **JavaScript (Web Speech API)**
  - **HTML/CSS**
  - **pyttsx3 (TTS)**
  - **Wikipedia API**
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## 9. Unique Selling Points (USP)

- Fully browser-based AI assistant
  - Voice + text hybrid interaction
  - No dependency on big ecosystems
  - Lightweight and fast
  - Easily extensible for future AI integrations
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## 10. Future Enhancements

- Integration with advanced AI models (LLMs)
- Mobile app version
- Smart home integration
- Personalized user profiles

- Multi-language support
  - AI memory & context awareness
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## 11. 🎯 Use Cases

- Students (study help, quick queries)
  - Developers (quick tools & commands)
  - General users (daily tasks automation)
  - Accessibility (voice-based computing)
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## 12. 🧠 Conclusion

Nova demonstrates how AI assistants can be:

- **Accessible**
- **Customizable**
- **Lightweight**

It serves as a strong foundation for building next-generation intelligent assistants with real-world applications.

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## 13. 👤 Author

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## ✅ Tip for Your VC Presentation

When presenting:

- Start with **problem → solution → demo flow**
- Show **live commands (voice if possible)**
- Emphasize **scalability + AI integration potential**